

Roberto Casadei^{, PhD}

CV (2026–10–04)

Associate Professor



Overview (highlights)

Current position: Associate Professor

2026/10 →

Department: Computer Science and Engineering (DISI)

Università di Bologna (IT)

Education: PhD in Computer Science and Engineering

2016/11 → 2020/04

Thesis: Engineering Self-* Collective Processes for Cyber-Physical Ecosystems

Università di Bologna (IT)

Research record & recent service:

Bibliometrics: 97+ in journals/conf.proceedings; H-index 27 (gscholar), 19 pubs in SCImago Q1 journals

Organisation: COORDINATION'26 PC Chair; AE Chair of iFM'25, ACSOS'24 & DisCoTec'23-25; PC member in 15+ conf. (6 A+ conf.); 5 workshops chaired

Editorial: EB member of Elsevier IoT, JAISCR, Wiley CCPE; Guest Ed. in Robotics&AI and SCP

Competitive projects

PI FIS3 Starting Grant for 1.1M€ – “Foundations for Macro-programming-based Sw Eng” (FoMaSE)

Teaching (recent/highlights). In total ≥597 hours and 9 different courses, including:

- Algorithms and Data Structures (44h, B. in Comp. Systems Technologies, UNIBO) a.y. 23-24 to 26-27
- Foundations of Informatics (30h, B. in Electrical/Biomedical Engineering) a.y. 20-21 to 25-26
- Software Design & Development (30h, B. in Comp. Systems Technologies, UNIBO) a.y. 22-23, 23-24

Supervision/Mentoring

- 2 PhD students (Gianluca Aguzzi, Nicolas Farabegoli), 1 researcher(s), 11 MEng + 10 BEng students

Mobility

- Visiting PhD: 2mo/2018 @ TU Wien (Prof. Dustdar), 3mo/2017 @ St Andrews Uni. (Prof. Dobson)

Selected awards: IEEE TCSC Award for Excellence'24

Awards: ACSOS'26 Best Artefact; ECOOP'22 Distinguished Artifact; TCSC PhD Dissertation Award'20

Recent Activity

Summary I am currently an Associate Professor at the Department of Computer Science & Engineering of the University of Bologna (UNIBO). I have 97+ publications at international journals and conferences; my current H-Index is 27 (GScholar), 23 (SCOPUS). I got a PhD in CS & Eng. from UNIBO, with a thesis awarded by the IEEE TCSC. I have been a visiting PhD student at the University of St Andrews (refereed by Prof. Simon Dobson) and at TU Wien (refereed by Prof. Schahram Dustdar). I got awards for research, service, and as a student. I received the IEEE TCSC Award for Excellence as early-career researcher. I have served as PC chair of COORDINATION'26 and editorial board member of Elsevier IoT, Wiley CCPE, JAISCR. I have been teaching 9+ different courses in Computer Science and Engineering Degrees since 2020, for more than 597 hours. I participate(d) to PRIN projects like *Fluidware* and *COMMON-WEARS* (Task Leader), and EU projects like DistriMuSe. I got the FIS3 Starting Grant for 1.1M€ for the “Foundations for Macro-programming Sw Eng” (FoMaSE) project.

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Research Themes in a Nutshell

In a nutshell, my research interests and activity revolve around two main themes: **software engineering** and **distributed artificial intelligence**. In particular, I focus on paradigms, models, and techniques fostering intelligence and autonomy in multi-agent and socio-technical systems. My research applies in the context of IoT-edge-cloud continua, large-scale CPSs, self-* systems, swarm robotics, and collective intelligence.

Qualifications

★ (ASN Fascia II) National Scientific qualification as associate professor in the Italian higher education system (Call 2021/2023, Ministerial Decree n. 553/2021 and 589/2021) for the disciplinary field of **01/B1 - Informatics** 2023-02 → 2033-02
I got this qualification by passing a CV evaluation by a committee on the basis of bibliometrics, publications, and qualifications.

★ (ASN Fascia II) National Scientific qualification as associate professor in the Italian higher education system (Call 2021/2023, Ministerial Decree n. 553/2021 and 589/2021) for the disciplinary field of **09/H1 - Information processing systems** 2022-02 → 2031-02
I got this qualification by passing a CV evaluation by a committee on the basis of bibliometrics, publications, and qualifications.

Qualification for teaching **Computer Science and Technologies** (c.c. **A041**) in Italian Secondary School 2021 →
I got this qualification by passing the ordinary public competition for STEM subjects in 2021 with a score of 192/200 (Rank #1).

Research Activity and Dissemination

Invited Seminars and Lectures

- **Invited seminar**: 2025-04-07 at *Gran Sasso Science Institute*, L'Aquila, Italy. Invited by Dr. Martina de Sanctis and Prof. Patrizio Pelliccione. Audience: PhD students, researchers, professors. Lecture title: "Software engineering methods for artificial collective intelligence".

Research Community Service and Participation in Committees

Summary

- **editorial board member** of 3 journals, including Elsevier IoT and JAISCR;
- **PC co-chair** for COORDINATION'26, and **Artefact track (co-)chair** for iFM'25, DisCoTec'23-24-25, ACSOS'24;
- PC member of 17 distinct conferences (including AAAI, ECAI, ACSOS), often for multiple editions, and 7 workshops;
- I covered most of the organisational roles in conferences: finance chair, publication/proceedings chair, track chair, web chair, publicity chair;
- reviewer for 16+ journals and 5+ books.

Membership in Editorial Boards

★ Elsevier *Internet of Things – Engineering Cyber Physical Human Systems* (Q1) - Editorial Board Member 2024–
<https://www.sciencedirect.com/journal/internet-of-things/about/editorial-board>
I have handled the editorial process for more than 24 submitted manuscripts.

- ★ **Wiley *Concurrency and Computation: Practice and Experience* (Q2) - Editorial Board Member** 2025–
<https://onlinelibrary.wiley.com/page/journal/15320634/homepage/editorialboard.html>
- ★ ***Journal of Artificial Intelligence and Soft Computing Research* (Q1) - Editorial Board Member** 2023–
<https://sciendo.com/journal/JAISCR?tab=editorial-board>
 JAISCR is an international journal, published by Sciendo (De Gruyter Open), publishing research on AI and soft computing.
Metrics: 2021 Journal Impact Factor: 2.675. SCImago Quartile 2022: Q1.
- Frontiers in Robotics and AI (Q2) – Multi-Robot Systems section – Review Editor** 2023–
<https://www.frontiersin.org/journals/robotics-and-ai/editors>

— Guest Editorial Roles

- Science of Computer Programming (Q3) - Software Track - Guest Editor** 2025–2026
- Special Issue “Selected Software Artifacts from the Papers of iFM 2025 - 20th International Conference on Integrated Formal Methods”
 I am guest editor for this Special Issue gathering software artifacts from high-quality submissions at iFM’25. Co-guest editors include Dr. Daniela Kaufmann.
Metrics: SCImago Quartile 2024: Q3.
- Science of Computer Programming (Q3) - Software Track - Managing Guest Editor - Special Issue “Selected Software Artifacts from the Papers of DisCoTec 2025 - 20th International Federated Conference on Distributed Computing Techniques”** 2025–2026
 I am the managing guest editor for this Special Issue gathering software artifacts from high-quality submissions at DisCoTec’25. Co-guest editors include Dr. Duncan Paul Attard, Dr. Vinicius Vielmo Cogo, and Dr. Emilio Incerto.
Metrics: SCImago Quartile 2025: Q3.
- Science of Computer Programming (Q2) - Software Track - Managing Guest Editor - Special Issue “Selected Software Artifacts from the Papers of DisCoTec 2024 - 19th International Federated Conference on Distributed Computing Techniques”** 2024–2025
 I am the managing guest editor for this Special Issue gathering software artifacts from high-quality submissions at DisCoTec’24. Co-guest editors include Dr. Duncan Paul Attard, Dr. Romyana Neykova, and Prof. Joao Soares.
Metrics: SCImago Quartile 2023: Q2.
- Science of Computer Programming (Q2) - Software Track - Managing Guest Editor - Special Issue “Selected Software Artifacts from the Papers of DisCoTec 2023 - 18th International Federated Conference on Distributed Computing Techniques”** 2023–2024
 I am the managing guest editor for this Special Issue gathering software artifacts from high-quality submissions at DisCoTec’23. Co-guest editors include Dr. Vinicius Vielmo Cogo, Dr. Tom van Dijk, and Prof. Alceste Scalas.
Metrics: SCImago Quartile 2023: Q2.

**Frontiers in Robotics & AI (Q2) - Guest Associate Editor - Special Issue
“Mobile Cyber-Physical Collectives”**

2021–2024

I am guest associate editor for a special issue, also called a research topic, entitled “Understanding and Engineering Cyber-Physical Collectives”. I have prepared, together with Prof. Lukas Esterle, the special issue proposal, contacted potential authors, run publicity campaigns, and been handling the review activity for submitted papers. Co-guest editors include Prof. Lukas Esterle, Prof. Rose Gamble, Dr. Paul Harvey, and Prof. Elizabeth F. Wanner. The special issue included 8 articles and 30 authors. The **editorial** is included in the paper section of this CV.

Metrics: SCImago Quartile 2021-23: Q2.

PC Chairing

COORDINATION'25 Program Committee Co-Chair

2026

[28th International Conference on Coordination Models and Languages](#) (part of the 21st International Federated Conference on Distributed Computing Techniques, DisCoTec'26).

Urbino, Italy

The event has been co-chaired with Dr. Fatemeh Ghassemi (University of Tehran).

Workshop Chairing

DISCOLI'25 Workshop Chair

2026

[5th DISCOLI Workshop on DIStributed COLlective Intelligence](#) (co-located with the 21st International Conference on Distributed Computing in Smart Systems and the Internet of Things, DCOSS-IoT'26).

Reykjavik, Iceland

The event has been co-chaired with Claudio Savaglio (University of Calabria, IT), Pietro Manzoni (Universitat Politècnica de València, ES), and Alessandro Sapienza (ISTC-CNR, IT).

DISCOLI'25 Workshop Chair

2025

[4th DISCOLI Workshop on DIStributed COLlective Intelligence](#) (co-located with the 21st International Conference on Distributed Computing in Smart Systems and the Internet of Things, DCOSS-IoT'25).

Lucca, Italy

The event has been co-chaired with Claudio Savaglio (University of Calabria, IT), Pietro Manzoni (Universitat Politècnica de València, ES), and Alessandro Sapienza (ISTC-CNR, IT).

ASMECC'23 Workshop Chair

2023

[1st ASMECC Workshop on Autonomic and Self-* Management for the Edge-Cloud Continuum](#) (co-located with the 4th IEEE International Conference on Autonomic Computing and Self-Organizing Systems - ACSOS 2023). The event has been being co-chaired with Stefano Forti (University of Pisa) and Lukas Esterle (Aarhus University).

Toronto, Canada

DISCOLI'23 Workshop Chair

[2nd DISCOLI Workshop on DIStributed COLlective Intelligence](#) (co-located with the 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things, DCOSS-IoT'23).

As Workshop Chair, I have prepared/submitted the workshop proposal, run publicity campaigns, selected the program committee, handled the peer review process, took the decisions regarding the final program, and chaired the workshop event. The event has been co-chaired with Noel Crespi (Institut Polytechnique de Paris), Claudio Savaglio (University of Calabria), and Christos Tsigkanos (University of Athens).

2023
Pafos, Cyprus

DISCOLI'22 Workshop Chair

[1st DISCOLI Workshop on DIStributed COLlective Intelligence](#) (co-located with the 42nd IEEE International Conference on Distributed Computing Systems, ICDCS'22).

As Workshop Chair, I have prepared/submitted the workshop proposal, run publicity campaigns, selected the program committee, handled the peer review process, took the decisions regarding the final program, and chaired the workshop event. The event has been co-chaired with Min Chen (Huazhong University of Science and Technology), Franco Zambonelli (University of Modena and Reggio Emilia), and Mengchu Zhou (New Jersey Institute of Technology).

2022
Bologna, Italy

eCAS'21 Workshop Chair

[6th eCAS Workshop on Engineering Collective Adaptive Systems](#) (co-located with ACSOS'21)

As Workshop Chair, I prepared/submitted the workshop proposal, run publicity campaigns, selected the program committee, handled the peer review process, and planned/run the workshop day. The event has been co-chaired with Lukas Esterle (Aarhus University).

2021
Washington DC, USA

eCAS'19 Workshop Chair

[4th eCAS Workshop on Engineering Collective Adaptive Systems](#) (co-located with SASO'19)

As Workshop Chair, I prepared/submitted the workshop proposal, run publicity campaigns, selected the program committee, handled the peer review process, and planned/run the workshop day. The event has been co-chaired with Soura Dasgupta (University of Iowa).

2019
Ulmea, Sweden

--- **Session chairing**

- PPDP'24, Session 3
- ACSOS'21, Session "Languages, formal methods, and assurances for Autonomic and Self-Organizing Systems"

--- **Participation in organising committees**

★ ACSOS'26 Finance Chair

7th IEEE International Conference on Autonomic Computing and Self-Organizing Systems

2025-26
Cesena, Italy

★ iFM'25 Artifact Evaluation Chair

[20th International Conference on Integrated Formal Methods](#)

2025
Paris, France

★ DisCoTec'25 Artifact Evaluation Chair

[20th International Federated Conference on Distributed Computing Techniques](#)

2025
Lille, France

★ ACSOS'24 Artifact Evaluation Chair 5th IEEE International Conference on Autonomic Computing and Self-Organizing Systems	2023-24 Aarhus, Denmark
★ DisCoTec'24 Artifact Evaluation Chair 19th International Federated Conference on Distributed Computing Techniques. Ditto.	2024 Groningen, Netherlands
CyberSciTech'23 Track Chair 8th IEEE Cyber Science and Technology Congress	2023 Abu Dhabi, UAE
★ DisCoTec'23 Artifact Evaluation Chair 18th International Federated Conference on Distributed Computing Techniques. Ditto.	2023 Lisbon, Portugal
CyberSciTech'22 Track 3 "Cyber Physical Computing & Systems" Chair 7th IEEE Cyber Science and Technology Congress I co-chair the Track on "Cyber Physical Computing & Systems".	2022 Calabria, Italy
★ ACSOS'22 Publication/Proceedings Chair 3rd IEEE International Conference on Autonomic Computing and Self-Organizing Systems	2021 Washington DC, USA
eCAS'22 Web Chair 7th eCAS Workshop on Engineering Collective Adaptive Systems	2022 Virtual
ACSOS'21 Publicity Chair 2nd IEEE International Conference on Autonomic Computing and Self-Organizing Systems	2021 Washington DC, USA
SASO'18 Web Chair 12th IEEE International Conference on Self-Adaptive and Self-Organizing Systems	2018 Trento, Italy

— Participation in program committees

Conferences:

1. **ACSOS** – IEEE International Conference on Autonomic Computing and Self-Organizing Systems
(2026, 2025 [Posters & Demos], 2024, 2023 [Senior PC], 2022, 2021 editions)
2. **SEAMS** – International Conference on Software Engineering for Adaptive and Self-Managing Systems
[ICORE'23: A]
(2026 edition)
3. **AAAI** – Annual AAAI Conference on Artificial Intelligence [ICORE'23: A*]
(2026, 2025, 2024, 2023 editions)
4. **ASE** – IEEE/ACM International Conference on Automated Software Engineering – New Ideas and Emerging Results (NIER) track [ICORE'23: A*]
(2026, 2024 edition)
5. **ECAI** – European Conference on Artificial Intelligence [ICORE'23: A]
(2025, 2024 edition)
6. **ECOOP/ISSTA** – European Conference on Object-Oriented Programming (ECOOP) / International Symposium on Software Testing and Analysis (ISSTA 2023) – ACM Student Research Competition (SRC)

[**ICORE'23**: A]
(2023 edition)

7. **MODELS** – ACM/IEEE International Conference on Model Driven Engineering Languages and Systems – ACM Student Research Competition (SRC) [**ICORE'23**: A]
(2021 edition)
8. **PLDI** – ACM SIGPLAN Conference on Programming Language Design and Implementation – Artefacts track [**ICORE'23**: A*]
(2021 edition)
9. **SAC** – ACM Symposium on Applied Computing – IRMAS “Intelligent Robotics and Multi-Agent Systems” track [**ICORE'23**: B]
(2026, 2025, 2024, 2023 editions)
10. **ICCCI** – International Conference on Computational Collective Intelligence [**ICORE'23**: B]
(2024, 2023 editions)
11. **CCNC** – IEEE Consumer Communications & Networking Conference [**ICORE'23**: B]
(2024 edition)
12. **COORDINATION** – International Conference on Coordination Models and Languages [**ICORE'23**: C]
(2026 [PC chair], 2025 [DisCoTec AEC chair], 2024 [DisCoTec AEC chair], 2023 [DisCoTec AEC chair], 2022, 2021 editions)
13. **ALIFE** – International Conference on Artificial Life [**ICORE'23**: C]
(2026, 2025 edition)
14. **ISCS** – International Symposium on Complex Systems
(2026 edition)
15. **QUATIC** – International Conference on the Quality of Information and Communication Technology – “Evolution and Maintenance” (2025) and “Quality in Adaptive Software” (2024) Tracks [**ICORE'23**: C]
(2025, 2024 editions)
16. **PPDP** – International Symposium on Principles and Practice of Declarative Programming [**ICORE'23**: C]
(2024 edition)
17. **DCOSS-IoT** – International Conference on Distributed Computing in Smart Systems and the Internet of Things [**ICORE'23**: B]
(2026, 2025 edition)
18. **CNA** – International Conference on Complex Networks and their Applications
(2025, 2024 editions)
19. **PDP** – Euromicro International Conference on Parallel, Distributed, and Network-Based Processing [**ICORE'23**: C]
(2025 edition)
20. **I3E** – IFIP Conference on e-Business, e-Services, and e-Society
(2025, 2024 editions)
21. **CyberSciTech** – IEEE Cyber Science and Technology Congress
(2023, 2022 editions)

Workshops

1. **EASI** – International Workshop on Engineering Autonomous Systems Intelligence
(<https://dat-workshop.github.io/> edition)
2. **DAT** – International Workshop on Digital Twin-enabled Autonomous Systems and Agents
(<https://dat-workshop.github.io/> edition)

3. **AHPC3** – International Workshop on Accelerated HPC in the Cloud-Edge Continuum
([2026 \(3rd\)](#), [2025 \(2nd\)](#), [2025 \(1st\)](#) editions)
4. **SMvg** – International Workshop on Smart Moving
([2024](#) edition)
5. **SISSY** – International Workshop on Self-Improving System Integration
([2025](#), [2024](#) editions)
6. **DISCOLI** – International Workshop on DIStributed COLlective Intelligence
([2024](#), [2023](#), [2022](#) editions)
7. **WOA** – International Workshop “From Objects to Agents”
([2024](#), [2023](#), [2022](#), [2021](#) editions)
8. **eCAS** – International Workshop on Engineering Collective Adaptive Systems
([2022](#), [2020](#), [2017](#) editions)
9. **ALPACA** – International Workshop on Adaptive, Learning PervAsive Applications
([2021](#) edition)

—— Peer review activity in international journals

- **Excellent reviewer** according to my [Web of Science™ profile](#) (according to the feedback provided by associate editors)
- 20+ verified reviews on my [Web of Science™ profile](#)
- ACM Transactions on Autonomous and Adaptive Systems ◦ IEEE Transactions on Intelligent Transportation Systems ◦ IEEE Internet of Things ◦ IEEE Transactions on Services Computing ◦ Elsevier Engineering Applications of Artificial Intelligence ◦ Elsevier Pervasive and Mobile Computing ◦ Elsevier Systems and Software ◦ Elsevier Science of Computer Programming ◦ Hindawi Scientific Programming ◦ MDPI Robotics ◦ MDPI Sensors ◦ MDPI Applied Sciences ◦ MDPI Informatics ◦ IOS Press Intelligenza Artificiale ◦ Springer Software and Systems Modeling ◦ Springer Autonomous Agents and Multi-Agent Systems ◦ World Scientific International Journal of Information Technology & Decision Making

—— Other review activity (books)

I am in the reviewer pool of *Manning Publications*, a well-known publisher of books on computer technology topics. The evidence of this activity lies in the “Acknowledgments” sections of the books that I reviewed. These include:

1. [Functional Design and Architecture](#) (Alexander Granin, 2024, Manning)
2. [Software Mistakes and Tradeoffs: How to make good programming decisions](#) (Tomasz Lelek and Jon Skeet, 2022, Manning)
3. [Quantum Computing in Action](#) (Johan Vos, 2022, Manning)
4. [Programming with Types](#) (Vlad Riscutia, 2019, Manning)
5. [Classic Computer Science Problems in Python](#) (David Kopec, 2019, Manning)

—— Talks at Conferences and Events

—— Presentations of peer-reviewed papers

Macroscopic Design of Swarms with Collective State Machines
Event 18th International Conference on Computational Collective Intelligence

2026/09
Crete, Grece

Macro-Programming Multi-Agent Systems: A Framework for Artificial Collective Intelligence (Blue Sky Ideas Track)	2026/05 Paphos, Cyprus / REMOTE
Event 25th International Conference on Autonomous Agents and Multiagent Systems, Blue Sky Ideas Track	
Integrating Collective Computing and the Social Internet of Things for Smart Cities: A Vision	2025/06 Lucca, IT
Event 4th DISCOLI Workshop on DIStributed COLlective Intelligence (DISCOLI 2025), co-located with DCOSS-IoT'25	
Declarative Macro-Programming of Collective Systems with Aggregate Computing: An Experience Report	2024/09 Milan, IT
Event 26th International Symposium on Principles and Practice of Declarative Programming, co-located with FM'24	
Software Engineering for Collective Cyber-Physical Ecosystems	2024/07 Porto de Galinhas, BR
Event International Workshop on Software Engineering in 2030, co-located with FSE'24	
Modelling Groups of Humans: Towards Crowd Digital Twins	2024/06 Paris, FR
Event 5th IEEE International Workshop on Social (Media) Sensing (SMS-24), co-located with IEEE ISCC'24	
Self-Organisation Programming: a Functional Reactive Macro Approach	2023/09 Toronto, CA
Event 4th IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2023)	
[Tutorial] Programming (and Learning) Self-Adaptive & Self-Organising Behaviour with ScaFi: for Swarms, Edge-Cloud Ecosystems, and More	2023/09 Toronto, CA
Event 4th IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2023)	
Programming Distributed Collective Processes for Dynamic Ensembles and Collective Tasks	2023/06 Lisbon, PT
Event 25th International Conference on Coordination Models and Languages (COORDINATION 2023)	
Towards Automated Engineering for Collective Adaptive Systems: Vision and Research Directions	2022/09 Falerna, ITA
Event 1st International Workshop on COMMunity-OrieNted WEARrable Computing Systems (COMMON-WEARS 2022)	
FScaFi: A Core Calculus for Collective Adaptive Systems Programming	2021/10 Rhodes, GRC
Event 10th International Symposium On Leveraging Applications of Formal Methods, Verification and Validation	
Augmented Collective Digital Twins for Self-Organising Cyber-Physical Systems	2021/10 Online
Event SISSY Workshop on Self-Improving System Integration	
Tuple-Based Coordination in Large-Scale Situated Systems	2021/06 Online
Event 23rd International Conference on Coordination Models and Languages	
Collective Adaptive Systems as Coordination Media: The Case of Tuples in Space-Time	2020/08 Online
Event 5th eCAS Workshop on Engineering Collective Adaptive Systems	

Engineering Resilient Collaborative Edge-enabled IoT	2019/07
Event 16th IEEE International Conference on Services Computing	Milan, ITA
Aggregate Processes in Field Calculus	2019/06
Event 21th International Conference on Coordination Models and Languages	Copenhagen, DNK
On Context-Oriented in Aggregate Programming	2019/06
Event 4th eCAS Workshop on Engineering Collective Adaptive Systems	Umeå, SWE
Coordinating Computation at the Edge: a Decentralized, Self-Organizing, Spatial Approach	2019/06
Event 4th IEEE International Conference on Fog and Mobile Edge Computing	Rome, ITA
Collective Abstractions & Platforms for Large-Scale Self-Adaptive IoT	2018/09
Event 3rd eCAS Workshop on Engineering Collective Adaptive Systems	Trento, ITA
From Field-Based Coordination to Aggregate Computing	2018/06
Event 20th International Conference on Coordination Models and Languages	Madrid, ESP
Compositional Blocks for Optimal Self-Healing Gradients	2017/09
Event 11th IEEE International Conference on Self-Adaptive and Self-Organizing Systems (SASO'17)	Tucson, AZ, USA
Practical Aggregate Programming in Scala	2016/10
Event Scala Symposium 2016	Amsterdam, NLD
Programming Actor-based Collective Adaptive Systems	2016/10
Event AGERE'16 (international workshop on agents and actors)	Amsterdam, NLD
On Execution Platforms for Large-Scale Aggregate Computing	2016/09
Event Workshop on Collective Adaptation in Very Large Scale Ubicomp: Towards a Superorganism of Wearables, Ubicomp/ISWC Adjunct	Heidelberg, DEU
Towards Aggregate Programming in Scala	2016/06
Event 1st International Workshop on Programming Models and Languages for Distributed Computing (PMLDC) – co-located with ECOOP	Rome, ITA

Other talks

Introduction to the 1st DISCOLI workshop on distributed collective intelligence	2022/06
Event 1st DISCOLI Workshop on DIStributed COLlective Intelligence	Bologna, IT
6th eCAS Workshop on Engineering Collective Adaptive Systems: Introduction to the workshop	2021/10
Event 6th eCAS Workshop on Engineering Collective Adaptive Systems	Online

Leadership and Supervision of Research Groups

- ★ **Leadership of the research group of the FIS3 Starting Grant project FoMaSE.** In this context, the following people have been supervised.
 - **Paolo Baldini**, Research Fellow (2026-2027, 1y), *Metodi della Complessità per Analisi e Progettazione Micro-Macro in Sistemi Computazionali (Complexity Science Methods for the Micro-Macro Analysis and Design in Computational Systems)*

Participations/Collaborations in Research Groups

- ★ **Leadership of the research group of the FIS3 Starting Grant project FoMaSE.** The team involves researchers hired on the project (**Dr. Paolo Baldini**). Other team members that contributed to the project include Roberto Girau, Mirko Viroli, Danilo Pianini, Angela Cortecchia (2026→)
- Collaboration with **Prof. Roberto Girau (University of Bologna, Italy)** on topics spanning robot swarms, Internet of Things, and digital twins, leading to 4+ co-authored publications (2024→)
- Collaboration with **Prof. Silvia Mirri and Dr. Giovanni Delnevo (University of Bologna, Italy)** on topics spanning digital twins and human-AI ecosystems, leading to 5+ co-authored publications (2024→)
- Participation in the research activity of the research group coordinated by **Prof. Mirko Viroli (University of Bologna, Italy)**, notably including **Prof. Danilo Pianini and Dr. Gianluca Aguzzi**. This is the research group with which Roberto has collaborated mostly in his career and that led to more than 60 publications in international journals and conference proceedings. (2016→)
- Collaboration with the research group coordinated by **Prof. Ferruccio Damiani (University of Turin, Italy)**, notably including **Prof. Giorgio Audrito and Prof. Gianluca Torta**, on themes related to field-based coordination and distributed computation. Output of this collaboration includes 22+ co-authored publication (of which 10+ articles on international journals). (2016→)
- Collaboration with the research group coordinated by **Prof. Giancarlo Fortino (University of Calabria, Italy)**, including notably **Prof. Clausio Savaglio** (6+ co-authored publications), on themes revolving around the software engineering of opportunistic services in the Internet of Things and edge computing. Output of this collaboration includes 5 co-authored publications (of which 3 articles published on international journals). The collaboration was also in the context of the [Fluidware project](#) and [COMMON-WEARS project](#). (2019→)
- Collaboration with **Danny Weyns (Katholieke Universiteit Leuven, Belgium)** on self-adaptive software architectures. Output of this collaboration includes 4 co-authored publications (of which 2 articles on international journals). (2020→2021)
- Collaboration with the research group coordinated by **Prof. Franco Zambonelli (University of Modena e Reggio Emilia, Italy; IEEE Fellow, ACM Distinguished Scientist)**, especially in the context of [Fluidware project](#). Output of this collaboration includes 6 co-authored publications (of which 2 articles on international journals). (2019→2023)
- Collaboration with **Prof. Lukas Esterle (University di Aarhus, Danimarca)**, on themes related to collective adaptive systems. Output of this collaboration includes one paper published in an international journal (ACM TAAS). We have also co-chaired the eCAS'21 workshop and co-edited one special issue. (2020→)
- Collaboration with **Prof. Alessandro Ricci (University di Bologna, Italy)**, on themes related to coordination and multi-agent systems. Output of this collaboration includes two papers published in the proceedings of international conferences. (2020→2021)
- Collaboration with **Prof. Guido Salvaneschi (University of St.Gallen)** on programming languages for distributed computing. Output of this collaboration includes 4 co-authored publications (of which 1 article published on an international journals). (2019→2024)
- Collaboration with **Prof. Volker Stolz (University of Oslo)** on themes related to distributed runtime verification and monitoring. Output of this collaboration includes 1 article published on an international journal. (2020→2021)
- Collaboration with **Prof. Simon Dobson (University of St Andrews)**, on themes related to sensor systems and complex networks. The collaboration included a 3-month visit period at the University of St Andrews. Output of this collaboration includes 1 article accepted on an international IEEE magazine. (2017→2020)

- Collaboration with the research group coordinated by **Prof. Schahram Dustdar (TU Wien, Austria; IEEE Fellow, ACM Distinguished Scientist)**, on themes related to the engineering of resilient, collaborative, Internet of Things systems. This collaboration started with my 2-month visit at TU Wien (Austria). Output of this collaboration includes 1 paper published in the proceedings of an international conference. (2018)
- Collaboration with **Prof. Antonio Bucchiarone (Fondazione Bruno Kessler, Trento, Italy)** on themes related to collective adaptive systems. Output of this collaboration includes 1 article accepted on an international IEEE magazine. (2019)
- Collaboration with **Prof. Alessandro Aldini (University of Urbino Carlo Bo', Italy)**, on the intersection of computational trust techniques and collective adaptive systems. Output of this collaboration includes 1 article published on an international journal and 1 paper in the proceedings of an international conference. (2017→2018).
- Collaboration with **Dr. Jacob Beal (Raytheon BBN Technologies, USA)** on aggregate computing. Output of this collaboration includes 1 article accepted on an international journal and 1 paper published in the proceedings of an international conference. (2018→2019)
- Collaboration with the research group coordinated by **Prof. Andrea Omicini (University of Bologna, Italy)**, on themes related to the engineering of multi-agent systems. Output of this collaboration includes 1 paper published in the proceedings of an international conference. (2018)

Participation in Funded Research Projects

- **Amount of funding obtained:** 1.105.342,03 EUR
 - 1.105.342,03 EUR (FIS3 Starting Grant, FoMaSE project)

Principal Investigator (PI)

- **FIS3 Starting Grant** for 1.1M€ – “Foundations for Macro-programming-based Sw Eng” (FoMaSE)
 - *FIS (Italian Science Fund)* is an ERC-inspired national funding program and competitive, peer-reviewed grant for individual researchers (PI), with emphasis on scientific excellence.
 - I ranked 3rd in area *PE6 - Computer Science and Informatics* with 47.5/50 pt with my 5-years-long project FoMaSE (Foundations for Macro-programming-based Software Engineering) and was granted 1.105.342,03 €.

Major Roles

- **Task Leader** for Task 1.2 “Collective Opportunistic Computing Model” within Work Package 1 “Model” in the PRIN Project “COMMON-WEARS” (2020HCWWLP).

Participation

- Participation as scientific collaborator in the regional project **DISCOV.ER** (2025-2026), led by Prof. Catia Prandi. <https://www.discover-project.it/>
- Participation as scientific collaborator in the EU-funded project **DistriMuSe – Distributed Multi-sensor Systems for Human Health and Safety** (HORIZON-KDT-JU-2023-2-RIA | 2023-2026). This project has received funding from the KDT Joint Undertaking, European Commission (grant agreement 101139769).
- Fixed-Term Researcher (RTD-A) in **Programma Operativo Nazionale (PON) “Research & Innovation” 2014-2020 – RTDA_GREEN project (CUP J41B21012140007)**.
- Participation as scientific collaborator in **PRIN Italian Project COMMunity-OrieNted WEARable Computing Systems (COMMON-WEARS, no. 2020HCWWLP)**, coordinated by Prof. Giancarlo Fortino and involving a consortium of four universities: University of Calabria, University of Bologna, University of Turin, and Rome Biomedical Campus University. I have participated to all the project meetings,

including the COMMON-WEARS Workshop at the 7th IEEE Cyber Science and Technology Congress. I have (co-)authored 2+ articles acknowledging the COMMON-WEARS project. [See the COMMON-WEARS website for information about the consortium, participants, and publications.](#) (2022→)

- Participation as scientific collaborator in **PRIN Italian Project Fluidware (2017KRC7KT)**, coordinated by Prof. Franco Zambonelli and involving a consortium of four universities: University of Modena and Reggio Emilia, University of Bologna, University of Camerino, and University of Calabria. The project revolves around models and techniques for adaptive distributed computing. I have participated to all the project meetings, and given various talks (such as [#1](#), [#2](#)). In particular, this collaboration activity led to two articles published in international journals and one paper on the proceedings of international conferences. In total, I have (co-)authored 8 articles acknowledging the FluidWare project. [See the FluidWare website for information about the consortium, participants, and publications.](#) (2019→2023)

Research for Public and Private Institutions

Research in Academia

- **Associate Professor** at the Department of Computer Science and Engineering, Alma Mater Studiorum - Università di Bologna, Italy). (2026-10→)
- **Fixed-Term Senior Researcher (RTD-B)** on project “Informatics” (Department of Computer Science and Engineering, Alma Mater Studiorum - Università di Bologna, Italy). (2023-10→ est. 2026-09)
- **Fixed-Term Junior Researcher (RTD-A)** on project “Techniques & strategies for Green Autonomic Internet of Things (GA-IoT)” (Department of Computer Science and Engineering, Alma Mater Studiorum - Università di Bologna, Italy), supervised by Prof. Mirko Viroli. (2022-02→2023-10)
- **Research fellowship (Assegno di Ricerca)** on project “Engineering evolving collective adaptive systems for smart infrastructures” (Department of Computer Science and Engineering, Alma Mater Studiorum - Università di Bologna, Italy), supervised by Prof. Mirko Viroli. (2021→2022)
- **Research fellowship (Assegno di Ricerca)** on project “Engineering evolving collective adaptive systems for modern infrastructures” (Department of Computer Science and Engineering, Alma Mater Studiorum - Università di Bologna, Italy), supervised by Prof. Mirko Viroli. (2020→2021)
- **Research fellowship (Assegno di Ricerca)** on project “Engineering collective adaptive processes through aggregate computing” (Department of Computer Science and Engineering, Alma Mater Studiorum - Università di Bologna, Italy), supervised by Prof. Mirko Viroli. (2019→2020)

Experiences Abroad

Visiting PhD student @ Technische Universität Wien (Austria)

2018, 2 months

Collaboration with the research group coordinated by *Prof. Schahram Dustdar (TU Wien, Austria; IEEE Fellow, ACM Distinguished Scientist)*, on themes related to the engineering of resilient, collaborative, Internet of Things systems. This collaboration started with my 2-month visit at TU Wien (Austria). Output of this collaboration includes one paper published in the proceedings of an international conference.

Visiting PhD student @ University of St Andrews (Scotland)

2017, 3 months

Collaboration with *Prof. Simon Dobson (University of St Andrews)*, on themes related to collective adaptive systems. Output of this collaboration includes one article accepted on an international IEEE magazine.

Marco Polo scholarship

Erasmus @ University of Limerick (Ireland)

2012, 4 months

Taking exams on specific BEng courses (see Education section).

Erasmus scholarship

Track Record

Scholarships and Grants

Computer Science and Engineering PhD Scholarship

2016/11 → 2019/10

Università di Bologna (IT)

This scholarship, granted by the Italian Ministry of Education, Universities, and Research (MIUR), covered all the 3 years of my PhD in Computer Science and Engineering at Alma Mater Studiorum–Università di Bologna (Italy). I got this scholarship by winning the corresponding public competition based on qualifications and interview.

Travel/Conference Grant – DisCoTec 2019

2019

Issuer Denmarks Tekniske Universitet (DTU)

This selective grant covered my participation to the COORDINATION 2019 conference in Copenhagen, Denmark.

Mobility Grant – Marco Polo 2016

2017

Issuer Department of Computer Science and Engineering (DISI), UNIBO

This selective grant covered my 3-months PhD abroad period in St Andrews, Scotland.

Travel/Conference Grant – Scala Symposium 2016

2016

Issuer École polytechnique fédérale de Lausanne (EPFL)

Grant issued by École polytechnique fédérale de Lausanne (EPFL) for presentation at the Scala Symposium 2016, co-located with the SPLASH'16 conference.

Study Scholarship (Borsa di Studio)

2016/09 → 2016/11

University of Bologna

Issuer Department of Computer Science and Engineering (DISI), UNIBO

Financ. Centro Studi - Orizzonti Holding SPA

Theme Software Infrastructures for the Management of IoT Systems

This scholarship supported a study of proof-of-concept IoT solutions for the retail market. In this context, I got familiarity with microservices, Docker, and process mining. I renounced to the scholarship once I got formally admitted to the PhD Programme, for incompatibility of the scholarships.

Awards

Research Awards

★ IEEE TCSC Award for Excellence 2024

2024/12

I was selected as winner for the Award for Excellence 2024 by the IEEE Technical Committee on Scalable Computing. The selection committee for the 2024 edition of the award was formed by Prof. Jinjun Chen (chair), Prof. Bernady O. Apduhan, Prof. Laurence T. Yang, Prof. Beniamino di Martino, and Dr. Eng. Didier El Baz.

*“The IEEE TCSC (Technical Committee on Scalable Computing) Award for Excellence in Scalable Computing (Early Career Researchers) recognizes up to 5 individuals who have made **outstanding, influential, and potentially long-lasting contributions in the field of scalable computing**. Typically the candidates are within 5 years of receiving their PhD degree as of January 01 of the year of the award.”*

The motivation of the award mentions: **“for contributions to computational and programming paradigms supporting the engineering of large-scale collective and self-organizing systems”**.

★ ECOOP 2022 Distinguished Artifact Award

Our artifact, associated to paper “Functional Programming for Distributed Systems with XC” and submitted to the Artifacts track of the 36th European Conference on Object-Oriented Programming (ECOOP 2022), has been awarded with the “Distinguished Artifact” award by the ECOOP’22 Artifact Evaluation Committee. Three artifacts were distinguished among a total of 53 artifacts accepted at ECOOP’22.

2022
Berlin (Germany)

★ IEEE TCSC Outstanding PhD Dissertation Award 2020

I was selected as winner for an international PhD thesis award promoted by the IEEE Technical Committee on Scalable Computing. The selection committee for the 2020 edition of the award was formed by Prof. Hai Jiang (chair), Prof. Bernady O. Aduhan, Prof. Beniamino di Martino, and Dr. Eng. Didier El Baz. “The IEEE TCSC Outstanding PhD Dissertation Award is an annual award to recognize candidates that have recently received a PhD degree for no more than 2 years and have written an **outstanding PhD dissertation in the field of the scalable computing** with applications. This award is established to encourage doctoral research that combines theory and practice or makes in-depth technical contributions, having the potential to contribute to the IEEE TCSC.”

2020

— Academic Service Awards

ACSOS’24 Certificate of Appreciation as “Artifact Evaluation Chair”

I was given this certificate of appreciation by by the General Chairs, Lukas Esterle and Alessandro V. Papadopoulos, and the PC Chairs, Payam Zahadat and Christian Becker, for my service as **Artifact Evaluation Chair** for the 5th IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2024).

2024

ACSOS’22 Outstanding Service Award

I was presented with the *Outstanding Service Award*, by the General Chairs Sven Tomforde and Kirstie Bellman, for having served as **Proceedings Chair** for the 3rd IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2022).

2022

ACSOS’21 Outstanding Service Award

I was presented with the *Outstanding Service Award*, by the General Chairs Jean Botev, Tarek El-Ghazawi, and Christopher Stewart, for having served as **Publicity Chair** for the 2nd IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2021).

2021

— Student Awards

Best District-2072 MEng graduate student (G.Bassi 2017 prize)

I was selected as winner of prize *Gianni Bassi*, with 5000€ scholarship, issued by Rotary Club Faenza for *Best Master Graduate Student* across all Engineering faculties of Romagna (district 2072) based on Grade Average and number of laudes.

2017
Faenza (Italy)

Prize for meritorious students (UNIBO)

I was selected among the ex-aequo winners of the public competition for meritorious students of the University of Bologna in academic year 2014/2015 (section common to all Engineering and Architecture faculties), granting a scholarship.

a.y. 2014-15
Università di Bologna (IT)

High school graduation award

I was given the “Roberto Ruffilli” acknowledgment for high-school graduation with score 100/100 (cum laude).

2009
Cesena (Italy)

Selected Publications

A selection of 15 significant publications follows (N.B.: metrics like IF are relative to the year of pub.).

1. **R. Casadei**, G. Aguzzi, G. Audrito, F. Damiani, D. Pianini, G. Scarso, G. Torta, and M. Viroli. “Software Engineering for Collective Cyber-Physical Ecosystems”. In: *ACM Transactions on Software Engineering and Methodology* (2025). DOI: [10.1145/3712004](https://doi.org/10.1145/3712004)
 - **Metrics:** SCImago Q1; IF 6.2.
 - **Short summary:** In this work, we provide a structure for research on collective cyber-physical ecosystems, and define a roadmap based on current software engineering challenges for such systems.
2. **R. Casadei**, G. Delnevo, R. Girau, and S. Mirri. “Crowd digital twins: Motivation, architecture, and roadmap”. In: *Future Generation Computer Systems* 182 (Sept. 2026), p. 108475. ISSN: 0167-739X. DOI: [10.1016/j.future.2026.108475](https://doi.org/10.1016/j.future.2026.108475)
 - **Metrics:** SCImago Q1; IF 6.1.
 - **Short summary:** This work proposes, analyses, and discusses the concept and implications of a “crowd digital twin”, i.e., a digital twin associated to physical gatherings of humans.
3. **R. Casadei**. “System-Wide IoT design and programming: Patterns for decentralised collective processes”. In: *Internet of Things* (2024), p. 101436. DOI: [10.1016/j.iot.2024.101436](https://doi.org/10.1016/j.iot.2024.101436)
 - **Metrics:** SCImago Q1; IF 7.6.
 - **Short summary:** In this work, I present a new design/programming pattern catalogue for collective computational processes. I propose a pattern catalogue distinguishing 5 Lifecycle patterns (Event-based triggering, Prolonged termination, Explicit collective termination, Spatial replication, Time replication), 2 Behaviour patterns (Leader-based and State-based Collective Behaviour), and 3 Mobility patterns (Node-attached, Space-attached, and Moving process).
4. **R. Casadei**, F. Fornari, S. Mariani, and C. Savaglio. “Programming IoT systems: A focused conceptual framework and survey of approaches”. In: *Internet of Things* (Feb. 2025), p. 101548. ISSN: 2542-6605. DOI: [10.1016/j.iot.2025.101548](https://doi.org/10.1016/j.iot.2025.101548)
 - **Metrics:** SCImago Q1; IF 7.6.
 - **Short summary:** This paper provides conceptual framework for analysing IoT programming approaches and a survey of state-of-the-art stream, event, micro and macro programming approaches for the IoT, identifying key open challenges for further research.
5. N. Farabegoli, D. Pianini, **R. Casadei**, and M. Viroli. “Dynamic IoT deployment reconfiguration: A global-level self-organisation approach”. In: *Internet of Things* 28 (Dec. 2024), p. 101412. ISSN: 2542-6605. DOI: [10.1016/j.iot.2024.101412](https://doi.org/10.1016/j.iot.2024.101412)
 - **Metrics:** SCImago Q1; IF 7.6.
 - **Short summary:** In this work, we investigate how “programs” expressing self-organisation logic by a global perspective can help to support the dynamic reconfiguration of IoT deployments. The contributions include a middleware for management of pulverised systems across the edge-cloud continuum examples of reconfiguring aggregate computations.

6. N. Farabegoli, D. Pianini, **R. Casadei**, and M. Viroli. “Scalability through Pulverisation: Declarative deployment reconfiguration at runtime”. In: *Future Generation Computer Systems* 161 (Dec. 2024), 545–558. ISSN: 0167-739X. DOI: [10.1016/j.future.2024.07.042](https://doi.org/10.1016/j.future.2024.07.042)
 - **Metrics:** SCImago Q1; IF 6.1.
 - **Short summary:** This work presents a framework for the reconfiguration of pulverised systems, a middleware architecture, and a practical implementation based on Kotlin-based DSLs (system DSL and deployment DSL).
7. G. Audrito, **R. Casadei**, F. Damiani, G. Salvaneschi, and M. Viroli. “The eXchange Calculus (XC): A functional programming language design for distributed collective systems”. In: *Journal of Systems and Software* (2024), p. 111976. ISSN: 0164-1212. DOI: <https://doi.org/10.1016/j.jss.2024.111976>
 - **Metrics:** SCImago Q1; IF 4.1.
 - **Short summary:** In this work, we present the eXchange Calculus (XC), a more general variant of the field calculus supporting differentiated messages and describe its support for distributed programming of resilient systems, with a narrative tailored to the programming language community.
8. **R. Casadei**. “Macroprogramming: Concepts, State of the Art, and Opportunities of Macroscopic Behaviour Modelling”. In: *ACM Comput. Surv.* 55.13s (2023), pp. 1–37. ISSN: 0360-0300. DOI: [10.1145/3579353](https://doi.org/10.1145/3579353)
 - **Metrics:** SCImago Q1; IF 16.
 - **Short summary:** This work provides the first comprehensive view of macroprogramming research. It is a long survey that provides a mapping study, a discussion of the key concepts and principles of the paradigm, a taxonomy of macroprogramming approaches, an analysis of more than 50 primary works on macroprogramming, and a discussion of related research opportunities and challenges.
9. G. Audrito, **R. Casadei**, and G. Torta. “A general framework and decentralised algorithms for collective computational processes”. In: *Future Generation Computer Systems* 158 (2024), pp. 11–27. ISSN: 0167-739X. DOI: [10.1016/j.future.2024.04.020](https://doi.org/10.1016/j.future.2024.04.020)
 - **Corresponding author**
 - **Metrics:** SCImago Q1; IF 7.
 - **Short summary:** This work provides a formalisation of collective/self-organising processes on event structures, with guarantees on their evolution dynamics. The approach is validated on a case study of peer-to-peer service discovery and access.
10. **R. Casadei**, G. Fortino, D. Pianini, A. Placuzzi, C. Savaglio, and M. Viroli. “A Methodology and Simulation-Based Toolchain for Estimating Deployment Performance of Smart Collective Services at the Edge”. In: *IEEE Internet of Things Journal* 9.20 (2022), pp. 20136–20148. DOI: [10.1109/JIOT.2022.3172470](https://doi.org/10.1109/JIOT.2022.3172470)
 - **Metrics:** SCImago Q1; IF 10.6.
 - **Short summary:** In this work, we present a methodology and toolchain for evaluating deployments of self-organizing systems developed through aggregate computing, across the IoT-fog-cloud continuum.
11. G. Audrito, **R. Casadei**, F. Damiani, V. Stolz, and M. Viroli. “Adaptive distributed monitors of spatial properties for cyber-physical systems”. In: *Journal of Systems and Software* 175 (2021), p. 110908. DOI: [10.1016/j.jss.2021.110908](https://doi.org/10.1016/j.jss.2021.110908)
 - **Metrics:** SCImago Q1; IF 3.514.
 - **Short summary:** In this work, we propose a compositional mapping of Spatial Logic for Closure Spaces (SLCS) constructs to field calculus constructs, enabling the direct encoding of SLCS formulas as decentralised monitors for runtime verification of spatial properties. We formally prove the generated monitors are correct and optimally self-stabilising, and validate the response to variable dynamics by means of simulations of crowd monitoring/control scenarios.

12. **R. Casadei**, M. Viroli, G. Audrito, D. Pianini, and F. Damiani. "Engineering collective intelligence at the edge with aggregate processes". In: *Engineering Applications of Artificial Intelligence* 97 (2021), p. 104081. ISSN: 0952-1976. DOI: [10.1016/j.engappai.2020.104081](https://doi.org/10.1016/j.engappai.2020.104081)
 - **Metrics:** SCImago Q1; IF 7.802.
 - **Short summary:** Aggregate processes are a programming abstraction, introduced as an extension to the field calculus and implemented in the ScaFi aggregate programming DSL, that captures *concurrent dynamic collective computations*. Its versatility in supporting collective intelligence and self-organisation is shown through simulations of IoT/edge computing scenarios.
13. D. Pianini, **R. Casadei**, M. Viroli, and A. Natali. "Partitioned integration and coordination via the self-organising coordination regions pattern". In: *Future Generation Computer Systems* 114 (Jan. 2021), pp. 44–68. DOI: [10.1016/j.future.2020.07.032](https://doi.org/10.1016/j.future.2020.07.032)
 - **Corresponding author**
 - **Metrics:** SCImago Q1; IF 7.307.
 - **Short summary:** This work presents a very general design pattern for decentralised feedback-regulated self-integration in dynamic environments. The *Self-organising Coordination Regions (SCR)* pattern consists of a dynamic distributed process involving leader election, coalition formation, and feedback loops between leaders and subordinates. The paper shows SCR has many known uses in literature and enjoys great versatility, shown via case studies in edge computing and hierarchical networks.
14. **R. Casadei**, G. Fortino, D. Pianini, W. Russo, C. Savaglio, and M. Viroli. "A development approach for collective opportunistic Edge-of-Things services". In: *Information Sciences* 498 (2019), pp. 154–169. DOI: [10.1016/j.ins.2019.05.058](https://doi.org/10.1016/j.ins.2019.05.058)
 - **Metrics:** SCImago Q1; IF 5.910.
 - **Short summary:** This work describes an approach to opportunistic edge computing that leverages collective-based services. In particular, it proposes a *Collective IoT Service* design abstraction. Experimental evaluation is performed through a crowd management case study, comparing Edge vs. Cloud deployments w.r.t. reactivity and precision.
15. **R. Casadei**, G. Fortino, D. Pianini, W. Russo, C. Savaglio, and M. Viroli. "Modelling and simulation of Opportunistic IoT Services with Aggregate Computing". In: *Future Generation Computer Systems* 91 (2018), pp. 252–262. DOI: [10.1016/j.future.2018.09.005](https://doi.org/10.1016/j.future.2018.09.005)
 - **Metrics:** SCImago Q1; IF 5.768.
 - **Short summary:** This work describes an aggregate approach of opportunistic computing for the IoT. The approach is based on the integration of the IoT Service Metamodel and the Aggregate Computing metamodel. Validation is performed with a simulated crowd safety case study.

Peer-Reviewed Publications and Bibliometrics

Bibliometrics

- **H-index:** 27 (Gscholar), 23 (Scopus). **i10-index:** 49 (Gscholar).
- **Number of citations:** 1924 (Gscholar), 1424 (Scopus).
- **Number of publications:** 97 (96 in Scopus, 82 in WoS).
- **Number of journal publications:** 37.
 - Number of Q1 journal publications: 19.
 - Number of Q2 journal publications: 11.
 - Number of Q3 journal publications: 7.

All my publications (ordered by Venue and Year DESC) follow.

Journal publications

1. **R. Casadei**, G. Delnevo, B. Bassi, C. Ceccarini, and S. Mirri. "AI-in-the-human-loop: an operational model for mature human-AI collaboration". In: *Journal of Ambient Intelligence and Humanized Computing* (Oct. 2026). ISSN: 1868-5145. DOI: [10.1007/s12652-026-05135-x](https://doi.org/10.1007/s12652-026-05135-x) 
2. **R. Casadei**, G. Delnevo, R. Girau, and S. Mirri. "Crowd digital twins: Motivation, architecture, and roadmap". In: *Future Generation Computer Systems* 182 (Sept. 2026), p. 108475. ISSN: 0167-739X. DOI: [10.1016/j.future.2026.108475](https://doi.org/10.1016/j.future.2026.108475) 
3. **R. Casadei**, D. P. Attard, E. D'Ossualdo, and R. Neykova. "Preface for the special issue on selected software artifacts from DisCoTec 2024 – The 19th international federated conference on distributed computing techniques". In: *Science of Computer Programming* 252 (July 2026), p. 103430. ISSN: 0167-6423. DOI: <https://doi.org/10.1016/j.scico.2025.103430> 
4. **R. Casadei**, G. Aguzzi, G. Audrito, F. Damiani, D. Pianini, G. Scarso, G. Torta, and M. Viroli. "Software Engineering for Collective Cyber-Physical Ecosystems". In: *ACM Transactions on Software Engineering and Methodology* (2025). DOI: [10.1145/3712004](https://doi.org/10.1145/3712004) 
5. A. Cortecchia, G. Ciatto, **R. Casadei**, and D. Pianini. "FieldVMC: an asynchronous model and platform for self-organising morphogenesis of artificial structures". In: *Complex & Intelligent Systems* (2025). ISSN: 2198-6053. DOI: [10.1007/s40747-025-02141-y](https://doi.org/10.1007/s40747-025-02141-y) 
6. G. Audrito, **R. Casadei**, F. Damiani, G. Torta, and M. Viroli. "Programming Distributed Collective Processes in the eXchange Calculus". In: *Logical Methods in Computer Science* Volume 21, Issue 4 (Oct. 2025). ISSN: 1860-5974. DOI: [10.46298/lmcs-21\(4:3\)2025](https://doi.org/10.46298/lmcs-21(4:3)2025) 
7. **R. Casadei**, F. Fornari, S. Mariani, and C. Savaglio. "Programming IoT systems: A focused conceptual framework and survey of approaches". In: *Internet of Things* (Feb. 2025), p. 101548. ISSN: 2542-6605. DOI: [10.1016/j.iot.2025.101548](https://doi.org/10.1016/j.iot.2025.101548) 
8. G. Aguzzi, **R. Casadei**, and M. Viroli. "MacroSwarm: A Field-based Compositional Framework for Swarm Programming". In: *Logical Methods in Computer Science* Volume 21, Issue 3 (Aug. 2025). ISSN: 1860-5974. DOI: [10.46298/lmcs-21\(3:13\)2025](https://doi.org/10.46298/lmcs-21(3:13)2025) 
9. **R. Casadei**. "System-Wide IoT design and programming: Patterns for decentralised collective processes". In: *Internet of Things* (2024), p. 101436. DOI: [10.1016/j.iot.2024.101436](https://doi.org/10.1016/j.iot.2024.101436) 
10. N. Farabegoli, D. Pianini, **R. Casadei**, and M. Viroli. "Dynamic IoT deployment reconfiguration: A global-level self-organisation approach". In: *Internet of Things* 28 (Dec. 2024), p. 101412. ISSN: 2542-6605. DOI: [10.1016/j.iot.2024.101412](https://doi.org/10.1016/j.iot.2024.101412) 
11. N. Farabegoli, D. Pianini, **R. Casadei**, and M. Viroli. "Scalability through Pulverisation: Declarative deployment reconfiguration at runtime". In: *Future Generation Computer Systems* 161 (Dec. 2024), 545–558. ISSN: 0167-739X. DOI: [10.1016/j.future.2024.07.042](https://doi.org/10.1016/j.future.2024.07.042) 
12. **R. Casadei**, L. Esterle, R. Gamble, P. Harvey, and E. F. Wanner. "Editorial: Understanding and engineering cyber-physical collectives". In: *Frontiers in Robotics and AI* 11 (2024). ISSN: 2296-9144. DOI: [10.3389/frobt.2024.1407421](https://doi.org/10.3389/frobt.2024.1407421) 
13. G. Audrito, **R. Casadei**, and G. Torta. "A general framework and decentralised algorithms for collective computational processes". In: *Future Generation Computer Systems* 158 (2024), pp. 11–27. ISSN: 0167-739X. DOI: [10.1016/j.future.2024.04.020](https://doi.org/10.1016/j.future.2024.04.020) 
14. G. Audrito, **R. Casadei**, F. Damiani, G. Salvaneschi, and M. Viroli. "The eXchange Calculus (XC): A functional programming language design for distributed collective systems". In: *Journal of Systems and*

Software (2024), p. 111976. ISSN: 0164-1212. DOI: <https://doi.org/10.1016/j.jss.2024.111976>

Q1

15. **R. Casadei**, V. Vielmo Cogo, T. van Dijk, and A. Scalas. “Preface for the special issue on selected software artifacts from DisCoTec 2023 – the 18th International Federated Conference on Distributed Computing Techniques”. In: *Science of Computer Programming* (Dec. 2024), p. 103255. ISSN: 0167-6423. DOI: [10.1016/j.scico.2024.103255](https://doi.org/10.1016/j.scico.2024.103255) Q3
16. **R. Casadei**, S. Mariani, D. Pianini, M. Viroli, and F. Zambonelli. “Space-Fluid Adaptive Sampling by Self-Organisation”. In: *Logical Methods in Computer Science* Volume 19, Issue 4 (Dec. 2023). ISSN: 1860-5974. DOI: [10.46298/lmcs-19\(4:29\)2023](https://doi.org/10.46298/lmcs-19(4:29)2023) Q2
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 77. G. Aguzzi, **R. Casadei**, D. Pianini, G. Salvaneschi, and M. Viroli. "Towards Pulverised Architectures for Collective Adaptive Systems through Multi-Tier Programming". In: *IEEE International Conference on Autonomic Computing and Self-Organizing Systems, ACSOS 2021, Companion Volume, Washington, DC, USA, September 27 - Oct. 1, 2021*. IEEE, 2021, pp. 99–104. DOI: [10.1109/ACSOS-C52956.2021.00033](https://doi.org/10.1109/ACSOS-C52956.2021.00033)
 78. G. Audrito, **R. Casadei**, and G. Torta. "Fostering resilient execution of multi-agent plans through self-organisation". In: *IEEE International Conference on Autonomic Computing and Self-Organizing Systems, ACSOS 2021, Companion Volume, Washington, DC, USA, September 27 - Oct. 1, 2021*. IEEE, 2021, pp. 81–86. DOI: [10.1109/ACSOS-C52956.2021.00076](https://doi.org/10.1109/ACSOS-C52956.2021.00076)
 79. **R. Casadei**, M. Viroli, G. Audrito, and F. Damiani. "FScaFi : A Core Calculus for Collective Adaptive Systems Programming". In: *Leveraging Applications of Formal Methods, Verification and Validation: Engineering Principles - 9th International Symposium on Leveraging Applications of Formal Methods, ISoLA 2020, Rhodes, Greece, October 20-30, 2020, Proceedings, Part II*. vol. 12477. Lecture Notes in Computer Science. Springer, 2020, pp. 344–360. DOI: [10.1007/978-3-030-61470-6_21](https://doi.org/10.1007/978-3-030-61470-6_21)
 80. **R. Casadei**, M. Viroli, and A. Ricci. "Collective Adaptive Systems as Coordination Media: The Case of Tuples in Space-Time". In: *2020 IEEE International Conference on Autonomic Computing and Self-Organizing Systems, ACSOS 2020, Companion Volume, Washington, DC, USA, August 17-21, 2020*. IEEE, 2020, pp. 139–144. DOI: [10.1109/ACSOS-C51401.2020.00045](https://doi.org/10.1109/ACSOS-C51401.2020.00045)
 81. **R. Casadei**, D. Pianini, G. Salvaneschi, and M. Viroli. "On Context-Orientation in Aggregate Programming". In: *IEEE 4th International Workshops on Foundations and Applications of Self* Systems, FAS*W@SASO/ICCAC 2019, Umea, Sweden, June 16-20, 2019*. IEEE, 2019, pp. 92–97. DOI: [10.1109/FAS-W.2019.00035](https://doi.org/10.1109/FAS-W.2019.00035)
 82. D. Pianini, **R. Casadei**, and M. Viroli. "Security in Collective Adaptive Systems: A Roadmap". In: *IEEE 4th International Workshops on Foundations and Applications of Self* Systems, FAS*W@SASO/ICCAC 2019, Umea, Sweden, June 16-20, 2019*. IEEE, 2019, pp. 86–91. DOI: [10.1109/FAS-W.2019.00034](https://doi.org/10.1109/FAS-W.2019.00034)
 83. S. Mariani, **R. Casadei**, F. Fornari, G. Fortino, D. Pianini, B. Re, W. Russo, C. Savaglio, M. Viroli, and F. Zambonelli. "Case Studies for a New IoT Programming Paradigm: Fluidware". In: *Proceedings of the 1st Workshop on Artificial Intelligence and Internet of Things*. Vol. 2502. CEUR Workshop Proceedings. CEUR-WS.org, 2019, pp. 82–96. URL: <http://ceur-ws.org/Vol-2502/paper6.pdf>
 84. **R. Casadei** and M. Viroli. "Collective Abstractions and Platforms for Large-Scale Self-Adaptive IoT". in: *2018 IEEE 3rd International Workshops on Foundations and Applications of Self* Systems (FAS*W), Trento, Italy, September 3-7, 2018*. IEEE, 2018, pp. 106–111. DOI: [10.1109/FAS-W.2018.00033](https://doi.org/10.1109/FAS-W.2018.00033)
 85. D. Pianini, G. Ciatto, **R. Casadei**, S. Mariani, M. Viroli, and A. Omicini. "Transparent Protection of Aggregate Computations from Byzantine Behaviours via Blockchain". In: *Proceedings of the 4th EAI International Conference on Smart Objects and Technologies for Social Good, GOODTECHS 2018, Bologna, Italy, November 28-30, 2018*. ACM, 2018, pp. 271–276. DOI: [10.1145/3284869.3284870](https://doi.org/10.1145/3284869.3284870)
 86. **R. Casadei**, A. Aldini, and M. Viroli. "Combining Trust and Aggregate Computing". In: *Software Engineering and Formal Methods - SEFM 2017 Collocated Workshops: FOCLASA, Trento, Italy, September 4-5, 2017, Revised Selected Papers*. Vol. 10729. Lecture Notes in Computer Science. Springer, 2017, pp. 507–522. DOI: [10.1007/978-3-319-74781-1_34](https://doi.org/10.1007/978-3-319-74781-1_34)
 87. G. Audrito, F. Damiani, M. Viroli, and **R. Casadei**. "Run-Time Management of Computation Domains in Field Calculus". In: *2016 IEEE 1st International Workshops on Foundations and Applications of*

Self Systems (FAS*W)*, Augsburg, Germany, September 12-16, 2016. IEEE, 2016, pp. 192–197. DOI: [10.1109/FAS-W.2016.50](https://doi.org/10.1109/FAS-W.2016.50)

88. M. Viroli, **R. Casadei**, and D. Pianini. “Simulating Large-scale Aggregate MASs with Alchemist and Scala”. In: *Proceedings of the 2016 Federated Conference on Computer Science and Information Systems, FedCSIS 2016, Gdańsk, Poland, September 11-14, 2016*. Vol. 8. Annals of Computer Science and Information Systems. IEEE, 2016, pp. 1495–1504. DOI: [10.15439/2016F407](https://doi.org/10.15439/2016F407)
89. M. Viroli, **R. Casadei**, and D. Pianini. “On Execution Platforms for Large-scale Aggregate Computing”. In: *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct*. UbiComp '16. Heidelberg, Germany: ACM, 2016, pp. 1321–1326. ISBN: 978-1-4503-4462-3. DOI: [10.1145/2968219.2979129](https://doi.org/10.1145/2968219.2979129)
90. **R. Casadei** and M. Viroli. “Towards Aggregate Programming in Scala”. In: *First Workshop on Programming Models and Languages for Distributed Computing, PMLDC@ECOOP 2016, Rome, Italy, July 17, 2016*. ACM, 2016, p. 5. DOI: [10.1145/2957319.2957372](https://doi.org/10.1145/2957319.2957372)

Contributions in books / collections

91. G. Aguzzi, **R. Casadei**, D. Pianini, and M. Viroli. “Self-organisation with Aggregate Computing: A Reflection Under the Lenses of Multi-agent Systems Engineering”. In: *The Agents Journey*. Springer Nature Switzerland, 2026, 147–178. ISBN: 9783032229403. DOI: [10.1007/978-3-032-22940-3_6](https://doi.org/10.1007/978-3-032-22940-3_6)
92. **R. Casadei**, F. Fornari, S. Mariani, and C. Savaglio. “Programming Approaches for Large-Scale IoT System Development: State of the Art”. In: *Fluidware: Novel Approaches for Large-Scale IoT Systems*. Ed. by F. Zambonelli, G. Fortino, B. Re, and M. Viroli. Cham: Springer International Publishing, 2024, pp. 21–45. ISBN: 978-3-031-62146-8. DOI: [10.1007/978-3-031-62146-8_2](https://doi.org/10.1007/978-3-031-62146-8_2)
93. **R. Casadei**, N. Farabegoli, G. Fortino, C. Savaglio, and M. Viroli. “Middleware Architectures for Fluid Computing”. In: *Fluidware: Novel Approaches for Large-Scale IoT Systems*. Ed. by F. Zambonelli, G. Fortino, B. Re, and M. Viroli. Cham: Springer International Publishing, 2024, pp. 49–63. ISBN: 978-3-031-62146-8. DOI: [10.1007/978-3-031-62146-8_3](https://doi.org/10.1007/978-3-031-62146-8_3)
94. D. Pianini, **R. Casadei**, S. Mariani, G. Aguzzi, M. Viroli, and F. Zambonelli. “Space-Fluid and Time-Fluid Programming”. In: *Fluidware: Novel Approaches for Large-Scale IoT Systems*. Ed. by F. Zambonelli, G. Fortino, B. Re, and M. Viroli. Cham: Springer International Publishing, 2024, pp. 107–134. ISBN: 978-3-031-62146-8. DOI: [10.1007/978-3-031-62146-8_6](https://doi.org/10.1007/978-3-031-62146-8_6)
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96. G. Aguzzi, **R. Casadei**, S. Mariani, M. Viroli, and F. Zambonelli. “Learning Opportunities in Collective Adaptive Systems”. In: *Fluidware: Novel Approaches for Large-Scale IoT Systems*. Ed. by F. Zambonelli, G. Fortino, B. Re, and M. Viroli. Cham: Springer International Publishing, 2024, pp. 179–199. ISBN: 978-3-031-62146-8. DOI: [10.1007/978-3-031-62146-8_10](https://doi.org/10.1007/978-3-031-62146-8_10)
97. **R. Casadei**, F. Damiani, G. Torta, and M. Viroli. “Actor-Based Designs for Distributed Self-organisation Programming”. In: *Active Object Languages: Current Research Trends*. Springer Nature Switzerland, 2024, 37–58. ISBN: 9783031510601. DOI: [10.1007/978-3-031-51060-1_2](https://doi.org/10.1007/978-3-031-51060-1_2). URL: http://dx.doi.org/10.1007/978-3-031-51060-1_2
98. **R. Casadei** and M. Viroli. “Programming Actor-Based Collective Adaptive Systems”. In: *Programming with Actors - State-of-the-Art and Research Perspectives*. Vol. 10789. Lecture Notes in Computer Science. Springer, 2018, pp. 94–122. DOI: [10.1007/978-3-030-00302-9_4](https://doi.org/10.1007/978-3-030-00302-9_4)

Edited volumes (curatele)

99. **R. Casadei** and F. Ghassemi, eds. *Coordination Models and Languages: 28th IFIP WG 6.1 International Conference, COORDINATION 2026, Held as Part of the 21st International Federated Conference on Distributed Computing Techniques, DisCoTec 2026, Urbino, Italy, June 8-12, 2026, Proceedings*. Springer Nature Switzerland, 2026. ISBN: 9783032283580. DOI: [10.1007/978-3-032-28358-0](https://doi.org/10.1007/978-3-032-28358-0). URL: <http://dx.doi.org/10.1007/978-3-032-28358-0>
100. **R. Casadei**, E. D. Nitto, I. Gerostathopoulos, D. Pianini, I. Dusparic, T. Wood, P. R. Nelson, E. Pournaras, N. Bencomo, S. Götz, C. Krupitzer, and C. Raibulet, eds. *IEEE International Conference on Autonomic Computing and Self-Organizing Systems, ACSOS 2022, Virtual, CA, USA, September 19-23, 2022*. IEEE, 2022. ISBN: 978-1-6654-7137-4. DOI: [10.1109/ACSOS55765.2022](https://doi.org/10.1109/ACSOS55765.2022)
101. **R. Casadei**, E. D. Nitto, I. Gerostathopoulos, D. Pianini, I. Dusparic, T. Wood, P. R. Nelson, E. Pournaras, N. Bencomo, S. Götz, C. Krupitzer, and C. Raibulet, eds. *IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion, ACSOS-C 2022, Virtual, CA, USA, September 19-23, 2022*. IEEE, 2022. ISBN: 978-1-6654-5142-0. DOI: [10.1109/ACSOS-C56246.2022](https://doi.org/10.1109/ACSOS-C56246.2022)

Artefacts

1. G. Aguzzi, **R. Casadei**, D. Pianini, and M. Viroli. *Dynamic decentralization domains for the Internet of Things - Simulation Repository*. 2022. DOI: [10.21227/QGBP-W789](https://doi.org/10.21227/QGBP-W789)
2. G. Audrito, **R. Casadei**, F. Damiani, G. Salvaneschi, and M. Viroli. "Functional Programming for Distributed Systems with XC (Artifact)". In: *Dagstuhl Artifacts Ser.* 8.2 (2022), 08:1–08:4. DOI: [10.4230/DARTS.8.2.8](https://doi.org/10.4230/DARTS.8.2.8)
3. **R. Casadei**, M. Viroli, G. Aguzzi, and D. Pianini. "ScaFi: A Scala DSL and Toolkit for Aggregate Programming". In: *SoftwareX* 20 (2022), p. 101248. ISSN: 2352-7110. DOI: <https://doi.org/10.1016/j.softx.2022.101248>
4. A. Placuzzi, D. Pianini, **R. Casadei**, M. Viroli, and D. Weyns. *Pulverization in Cyber-Physical Systems: Engineering the Self-Organizing Logic Separated from Deployment (Artifact)*. aPlacuzzi/Experiment-2020-FutureInternet-LoRa (Version 0.1.0-2020-11-24T153641). 2020. DOI: [10.5281/ZENODO.4288898](https://doi.org/10.5281/ZENODO.4288898)

Contributions (e.g. open-source/academic software projects)

SCAFI (Project Lead and Developer)

ScaFi is a Scala toolkit for Aggregate Computing: it includes a field calculus DSL, simulator, distributed platform, and more. It is a collection of project modules for a total of more than 140K LoC. I lead the project and development of ScaFi, check pull requests, and coordinate work.

ALCHEMIST (Contributor)

Alchemist is a meta-simulator tailored to pervasive computing, on the JVM. I issued pull requests mainly on the ScaFi-Alchemist incarnation.

Miscellaneous Contributions

– org.protelis.protelisdok: Kotlin gradle plugin to generate Protelis docs, via dokka

Slides @ Slideshare

Presentations corresponding to lecture notes (mainly about computer science) and conference talks.

Certifications

TS: Microsoft .NET Framework - Application Development Foundation
(License 8424975 - Prometric)

2011
Bologna (Italy)

Teaching and Supervision

Teaching

Aggregate statistics (2020-now):

- **Invited lectures:** 1
- **Types of courses taught (despite editions):** 2 PhD courses; 1 MOOCs; 0 Master courses; 1 CAF courses; 5 Bachelor courses; 2 IFTS courses
- **Total number of hours of teaching (PhD/M./B./CAF):** 597
- **Total number of taught course modules:** 19 (15 with teaching responsibility)
- **Total number of different taught courses:** 9 ([CAF] Complex Networks, [PhD] Collective Intelligence, [PhD] SLR, [B.] Foundations of Informatics, [B.] Software Design and Development, [B.] Algorithms and Data Structures, [B.] Mobile System Programming, [B.] Object-Oriented Programming)
- **Total number of hours in other teaching-related activities:** 300 (tutoring), 90 (IFTS)

Invited lectures

- **Invited lecture:** 2025-05-05 to 2025-05-09 (5 days, 2 lectures, 6 hours) at Kaunas Technical University, Kaunas, Lithuania. Audience: Bachelor's degree students. Topic: software engineering. Lecture topic: decentralised version control with git.

Massive Online Open Courses (MOOCs)

- **Python tools for Data Analysis and Machine Learning:** video lectures, supplementary material, and quizzes.
 - **What:** This MOOC was produced as part of the **Edvance project— Digital Education Hub per la Cultura Digitale Avanzata**. The project is funded by the European Union – Next Generation EU, Component 1, Investment 3.4 “Didattica e competenze universitarie avanzate”
 - **Language:** English. **Hours of activity:** 9. **EQF Level:** 6 (Bachelor's). **ISCED-F:** 0613 Software and applications development and analysis.

Teaching in PhD and Higher Education courses

Literature reviews: methods and tools (phd course)

Course PhD Programme on Computer Science and Engineering, 41st Cycle
Activity I run a module of 12 hours about methods and tools for planning and running (systematic) literature reviews.

2026
Cesena, UNIBO

Big Data and Machine Learning

Course 6036 - Corso di Alta Formazione (CaF) in Transizione digitale nella logistica portuale
Activity 5 hours of lectures on complex systems and complex networks, with examples in Python.

a.y. 2023-24
Ravenna, UNIBO

Engineering Intelligent Collective Systems (phd course)

Course PhD Programme on Computer Science and Engineering
Activity I run a module of 10 hours on research themes pertaining to collective adaptive systems engineering.

2021
Cesena, UNIBO

— Teaching in BEng/BSc courses

09730 - Principles of Informatics - 6 cfu

a.y. 2025-26

15305 - Foundations of Informatics A - 6 cfu

Cesena, UNIBO

Role I got the **responsibility** for the entire course.

Course First cycle degree programme (L) in Electronics Engineering and in Biomedical Engineering

Activity I run 3 CFUs (30 hours) of teaching and laboratory material preparation and assessment on the fundamentals of computers science (representation of information, algorithms) and the basics of imperative, structured programming in C.

11929 - Algorithm and Data Structures - 6 cfu

a.y. 2025-26

Role I got the **responsibility** for the entire course.

Cesena, UNIBO

Degree First cycle degree programme (L) in Computer Systems Technologies (cod. 6007)

Activity I teach (for 48 hours) algorithm analysis and design, and data structures, with examples in pseudocode, C, and Python.

11929 - Algorithm and Data Structures - 6 cfu

a.y. 2024-25

(see edition above)

Cesena, UNIBO

09730 - Principles of Informatics - 6 cfu

a.y. 2024-25

15305 - Foundations of Informatics A - 6 cfu

Cesena, UNIBO

(see edition above)

95648 - Software Design and Development - 9 cfu

a.y. 2023-24

Role I got the **responsibility** for the entire course.

Cesena, UNIBO

Course First cycle degree programme (L) in Computer Systems Technologies

Activity I teach (for 30 hours) the basics of software design and development, object-oriented programming, and Java.

11929 - Algorithm and Data Structures - 6 cfu

a.y. 2023-24

(see edition above)

Cesena, UNIBO

B3109 - Mobile System Programming Workshop Classes - 6 cfu

a.y. 2023-24

Role I got the **responsibility** for the entire course.

Imola, UNIBO

Course First cycle degree programme (L) in Computer Systems Technologies (cod. 6007 and cod. 5816)

Activity I teach (for 16 hours) mobile application development, with a focus on the Android platform and Kotlin programming.

09730 - Principles of Informatics - 6 cfu

a.y. 2022-23

15305 - Foundations of Informatics A - 6 cfu

Cesena, UNIBO

(see edition above)

95648 - Software Design and Development - 9 cfu

a.y. 2022-23

(see edition above)

Cesena, UNIBO

70219 - Object-Oriented Programming - Module 3

a.y. 2022-23

Course First cycle degree programme (L) in Computer Science and Engineering

Cesena, UNIBO

Activity I teach (for 30 hours) OOP in Java and related programming tools (version control, Eclipse, JavaFX, Gradle, etc.).

98214 - Intelligent Cyber-Physical Systems - Module 2

Course Minor "Smart Infrastructures"

Activity I teach (for 20 hours) about topics related to intelligent cyber-physical systems, including tutoring for projects.

a.y. 2021-22
Cesena, UNIBO

09730 - Principles of Informatics – 9 cfu

(see edition above)

a.y. 2021-22
Cesena, UNIBO

70219 - Object-Oriented Programming - Module 3

(see edition above)

a.y. 2021-22
Cesena, UNIBO

15305 - Foundations of Informatics A - Module 2

Course First cycle degree programme (L) in Electronics Engineering for Energy and Information (cod. 8767)

Activity I teach (for 30 hours) the fundamentals of computers science (representation of information, algorithms) and the basics of imperative, structured programming in C.

a.y. 2020-21
Cesena, UNIBO

70219 - Object-Oriented Programming - Module 3

(see edition above)

a.y. 2020-21
Cesena, UNIBO

— Other teaching-related activities

Tutor for course modules *Programming and Development Paradigms and Concurrent and Distributed Programming* (five editions)

Course Second cycle degree programme (LM) in Computer Science and Engineering (cod. 8614)

Activity As a tutor, I prepare exercises, help students in doing them and following the course, and do seminars on specialised topics. These courses cover advanced programming and paradigms (functional, logic, concurrent, distributed). The contract was of 40 hours (2016-17), 60 hours (2017-18), 60 hours (2018-19), 24 hours (2019-20), 60 hours (2020-21).

a.y. 2016-17, 2017-18,
2018-19, 2019-20, 2020-21
Cesena, UNIBO

Tutor for course module *Object-Oriented Programming*

Course First cycle degree programme (L) in Computer Science and Engineering

Activity As a tutor, I prepare exercises (on OOP in Java) and help students in doing them and following the course. The contract is for 56 hours.

a.y. 2019-20
Cesena, UNIBO

Seminars in the BBS open-program *Internet of Things*, modules *Software Production and Distributed Systems*

Activity I did two seminars to engineers and professionals on specialised topics: (1) software testing and (2) cloud-native application development.

2018
Bologna Business School

30h in IFTS course "Technician for design and development of computer applications specialised in new digital technologies"

What Higher Technical Education and Training (IFTS) is an education programme funded by the Italian Ministry of Education (MIUR).

Activity 30 hours teaching the basics of programming in Python to high-school graduates.

2018
Cesena

60h in IFTS course "Technician for design and development of computer applications specialised in business problem solving"

Activity 60 hours teaching the basics of programming in JavaScript to high-school graduates.

2018
Cesena

Supervision and Mentoring of Students / Researchers

Aggregate statistics:

- PhD students mentored: 2
- Research fellows mentored: 1
- MEng/Msc student thesis (co-)supervision: 11
- BEng/Bsc student thesis (co-)supervision: 13
- BSc student internship supervision: 7

Full list of (co-)supervised theses available in amslaurea.unibo.it.

PhD Students in Computer Science and Engineering

1. [Gianluca Aguzzi](#) (post-doc, former PhD student): I have been a de-facto supervisor of Gianluca during his PhD. Gianluca's research has focussed on aggregate programming and multi-agent reinforcement learning. The mentoring is also witnessed by [17 co-authored publications out of the total of 23 publications as of 2024-07](#) of his PhD (2021–2024).
2. [Nicolas Farabegoli](#) (PhD student): I have been mentoring Nicolas during his PhD. Nicolas' research has focussed on distributed middlewares and deployment platforms for collective computing systems. The mentoring is also witnessed by [5 co-authored publications during his PhD \(2023–2026\)](#).

Researchers

1. [Andrea Placuzzi](#) (1-year research contract). I have been a de-facto supervisor of Andrea. His research activity has focussed on simulation and deployment techniques for collective systems. The mentoring is also witnessed by [3 co-authored publications out of the total of 3 publications](#) of his research activity.

M.Sc. / M.Eng. Students in Computer Science and Engineering

1. [Event-driven simulation and verification of FRASP systems against spatio-temporal properties](#), Jahrim Gabriele Cesario, 2024
2. [Gestione degli effetti in linguaggi di programmazione funzionale: tecniche di modellazione e interpretazione](#), Giacomo Cavalieri, 2023
3. [A functional-reactive perspective on the Aggregate Computing paradigm](#), Francesco Dente, 2023
4. [A platform for aggregate computing over LoRaWAN network](#), Andrea Placuzzi, 2020
5. [A Reinforcement Learning approach to discriminate unsafe devices in aggregate computing systems](#), Chiara Volonnino, 2020
6. [Una piattaforma client-server universale per Aggregate Computing](#), Loris Cangini, 2020
7. [Towards Aggregate Processes in a Field Calculus-Based Platform](#), Davide Foschi, 2018
8. [Distributing Aggregate Computations on top of Akka Actors](#), Manuel Peruzzi, 2018
9. [Prototyping a scalable Aggregate Computing cluster with open-source solutions](#), Cristian Paolucci, 2018
10. [Tecniche e algoritmi di aggregate computing a supporto di contesti di smart mobility](#), Filippo Berlini, 2017
11. [Design and Deployment of an Execution Platform based on Microservices for Aggregate Computing in the Cloud](#), Thomas Farneti, 2017

B.Sc. / B.Eng. Students in Computer Science and Engineering

1. [Gestione di utenti multi-applicazione in un sistema di Single Sign-On a microservizi](#), Giacomo Ghinelli

2. *Implementazioni di soluzioni Unified Communication & Collaboration in ambienti on-premise e in cloud*, Alessandro Rambaldi
3. *Progettazione e sviluppo di un sistema di gestione automatizzato degli stati e degli accessi veicolari*, Yuhang Zhu
4. *Progettazione e Sviluppo di un Sistema di Versionamento in Backend SaaS Multitenant*, Gabriele Broccoli, 2025
5. *Sviluppo Progetto Cloud Nativo Ibrido per la Pubblica Amministrazione*, Alessandro Agostini, 2024
6. *Progettazione e sviluppo di una web app per la somministrazione di questionari*, Luca Nicolae Ferar Tofan, 2024
7. *Un'indagine sugli strumenti di supporto alle revisioni sistematiche della letteratura scientifica*, Edoardo Montanari, 2023
8. *Progettazione di un ambiente di programmazione visuale block-based per ScaFi*, Matteo Cerioni, 2022
9. *Un framework per la graficazione di dati in Scala*, Andrea Bianchi, 2022
10. *Piattaforma a Supporto del Monitoraggio di Sistemi di Computazione Aggregata: Caso di Studio ScaFi-Web*, Denys Grushchak, 2021
11. *Progettazione di un sistema di categorizzazione delle regressioni per il compilatore Rust*, Giacomo Pasini, 2020
12. *Design e prototipazione di un middleware per applicazioni aggregate location-based*, Linda Vitali, 2020
13. *Sviluppo di una libreria in Scala di supporto alla creazione e configurazione di uno stack SMACK*, Stefano Salvatori, 2018
14. *Sviluppo di un front-end di simulazione per applicazioni aggregate nel framework Scafi*, Gianluca Aguzzi, 2018
15. *Sviluppo di applicazioni distribuite con lo stack SMACK*, Emiliano Ciavatta, 2018
16. *Indagine sull'utilizzo di Scala per progetti Android*, Giuseppe Ettore Radaelli, 2017

Roles, Responsibilities, Institutional, and Extra-Institutional Activity

Expert evaluator for research and innovation funding agencies

- Expert innovation evaluator within the “**Albo esperti innovazione tecnologica**” del MIMIT (Ministero delle Imprese e del Made in Italy) (2025-2026)
- Expert reviewer for **Leverhulme Trust** (2024)

National Activity

- Substitute Member of the Judging Committee (① *Commissione Giudicatrice*) for the Thesis Defence of Lorenzo Testa, University of Turin, PhD in INFORMATICA (XXXVI cycle)

Participation in University Boards and Committees

PhD Programme Boards

- Member of the Board of the inter-departmental PhD Programme (① *Collegio di Dottorato*) in **Engineering and Information Technology for Structural and Environmental Monitoring and Risk Management (EIT4SEMM)**, University of Bologna

- Member of the Board for PhD Students enrolled to the PhD Programme in Computer Science and Engineering, University of Bologna
 - Ruslan Shaiakhmetov (2023–2025)
 - Martina Baiardi (2024–2026)

Department Boards

- Member of the **Web Board** ( *Commissione Web*) (since 2025-02) of the DISI Department: the activity involves the coordination of the web-based communication strategy of the DISI department.
- Member of the **Orientation Board** ( *Commissione Orientamento*): this involves the orientation activities aimed at secondary school students (e.g., presentations of the Degree courses during events like AlmaOrienta).
- Member of the Board for the Selection of a Research Contract ( *Assegno di ricerca*) for project “Study, implementation, and validation of a prototype for an interconnected Digital Platform for remote software update of industrial machines” ( “*Studio, realizzazione, e validazione di un prototipo di Piattaforma Digitale interconnessa per l’aggiornamento del software da remoto per macchine industriali*”)

Course Degree Boards

- Member of the Board for the **Quality Assurance (QA)** and Review ( *Riesame*)(since 2023-10), First Cycle Degree/Bachelor in Computer Systems Technologies (Qualifying Vocational Degree Programme), University of Bologna
- President (since 2024-9) and Member (since 2023-10) of the Board for the **Apprenticeships** ( *Commissione Tirocini*), First Cycle Degree/Bachelor in Computer Systems Technologies (Qualifying Vocational Degree Programme), University of Bologna
- Member of the Graduation Board ( *Sottocommissione di Laurea*) (2023-07-17, 2022-11-30, 2022-05-25), First Cycle Degree/Bachelor in Computer Science and Engineering, University of Bologna
- Member of the Board for the “Admission to degree years following the first one” ( *Bando per l’ammissione ad anni successivi al primo*) (a.y. 24-25, 23-24), First Cycle Degree/Bachelor in Computer Systems Technologies (Qualifying Vocational Degree Programme), University of Bologna
- Member of the Board for the “Extraordinary Selection” (2022-09-19), First Cycle Degree/Bachelor in Computer Systems Technologies (Qualifying Vocational Degree Programme), University of Bologna
- Member of the Board for the “Competition Test for the Admission to the Degree Course” (2022-07-21, 2022-07-19), First Cycle Degree/Bachelor in Computer Systems Technologies (Qualifying Vocational Degree Programme), University of Bologna
- TOLC (Test OnLine CISIA) committee for University Admission (2024-03-28), University of Bologna

Memberships and Participation in Organisations and Societies

- ★ **IEEE Senior Member** I have been elevated to the IEEE Senior Member grade in July, 2025. IEEE Senior Membership is an honor bestowed only to those who have made significant contributions to the profession. This requires 10 years of professional experience, 5 years of significant performance, and referees by at least 3 IEEE members holding the Fellow, Senior Member, or Honorary Member grade. The evaluation is performed by the IEEE Admission & Advancement Committee. 2025-
- IEEE Member** Including membership to Computer Society Technical Community on Scalable Computing (5y) 2017-2025

ACM Member

2017-2024

CIRI ICT member Participation to the “Interdepartmental Centre for Industrial Research” (CIRI) on ICT (Information and Communication Technology)

2025-

GRIN Società Informatica Italiana - Member

2024–

IEEE Technical Committee on Computational Collective Intelligence - Member

2022–

Third Mission

Events, school, and society

- Lecture on “Collective intelligence: from Nature to Robots” – delivered in the context of **Unijunior Romagna 3rd edition** (2026-05)
 - I **contributed** to the conceptualisation of the lecture.
- **Project Emerge**, a demo for the **Researchers' Night 2025**
 - I **contributed** to the conceptualisation of the project. The project builds on a programming language that I developed and led: ScaFi.
- **Forum Young Campus**, Cesena Campus, 2026-05-07
 - I participated as a **speaker** in this event. The guiding theme was the “Journey”, understood as a personal and collective process passing through education, work, technology, life choices, failures, and transformations. The event was based on round tables involving academics and entrepreneurs discussing experiences and trends.

Projects with Companies

- Participation as scientific collaborator in project on “cybersecurity and threat attribution” by YOROI S.R.L. and University of Bologna. (2021→2022)
- Participation as scientific collaborator in project “Realizzazione di un Compilatore da specifica di alto livello a comandi PLC per macchine monolama per la lavorazione del legno” by the University of Bologna and SCM Group S.p.A. (2020→2021)
- Participation as scientific collaborator in project “Realizzazione tramite Model Driven Engineering di un sistema di reporting moderno per l'ERP Star4” by the University of Bologna and Formula Impresoft S.R.L. (2020→2021)
- Participation as scientific collaborator in project PG/2016/667492 “Re-ingegnerizzazione da Cobol a .NET di una piattaforma gestionale intersettoriale” by University of Bologna and Harvard Group (now Impresoft S.R.L.). (2017→2018)
- Research activity “Software infrastructures for the management of IoT systems” on scholarship issued by the Department of Computer Science and Engineering (Alma Mater Studiorum - Università di Bologna), on funds by Centro Studi - Orizzonti Holding. The research focussed on technologies for IoT, process mining, and microservices, and their application to large-scale and small-scale retail. (2016→2017)

Background and miscellaneous

Professional Experience

Full-Stack Software Engineer

Web service (WCF) and application development in ASP.NET MVC and JavaScript within a Scrum/Kanban process framework.

2014/09 → 2015/12

[Apex-Net \(WEDO\)](#)

Cesena (Italy)

- Server-side: ASP.NET MVC
- Client-side: ZURB Foundation, HTML, CSS3, JavaScript, JQuery, KnockoutJS
- Client-server interaction: Comet via SignalR
- Frameworks/libs: DevExpress ASP.NET MVC Extensions
- Web services: WCF (REST-style, SOAP XML & WS-*)

Mobile Software Engineer

Development of a Windows 8.1 application in C#/XAML and development of the related WCF back-end service for SharePoint integration.

2014/03 → 2014/09

[Apex-Net \(WEDO\)](#)

Cesena (Italy)

IT Book reviewer

During the years of high school, I used to write reviews of computer science books for an Italian e-zine (and the next years, more informally, for a [personal blog](#)) The website is dismissed: [look it up on archive.org](#).

2007 → 2011

[programmazione.it](#)

Education

PhD Programme in Computer Science and Engineering

2016/11/01 → 2020/04/02

Thesis [Engineering self-adaptive collective processes for cyber-physical ecosystems](#) Università di Bologna (IT)

- Courses**
- [Spatial Multiagent Systems and Aggregate Computing: New Directions for Spatial Computing](#) (2017, A. Omicini & M. Viroli)
 - [Approximation Algorithms](#) (BISS'17, F. Grandoni)
 - [Kleene Algebra with Tests and Applications to Network Programming](#) (BISS'17, A. Silva)
 - [Models and Algorithms for Matching and Assignment Problems](#) (S. Martello)
 - [Developing, maintaining, and sharing software tools for research](#) (D. Pianini)

24 CFU Training Programme – Anthropological, psycho-pedagogy disciplines and teaching methodologies and technologies

2018

Università di Bologna (IT)

- Exams**
- Anthropology, 30L/30
 - Psychology, 30/30
 - Pedagogy, special pedagogy, and didactics for inclusion, 30/30
 - General methodologies and technologies for didactics, 30L/30

Master's Degree in Computer Science and Engineering

2013/09 → 2016/03

Grade Summa Cum Laude (Grade Average: 30/30, 6 laudes)

Università di Bologna (IT)

Thesis [Aggregate Programming in Scala: a Core Library and Actor-based Platform for Distributed Computational Fields](#) (supervisor: Mirko Viroli)

Exams

- Artificial Intelligence (Vittorio Maniezzo), 30/30
- Programming and Paradigms (Alessandro Ricci), 30L/30
- Autonomous Systems (Andrea Omicini), 30L/30
- Business Intelligence (Stefano Rizzi), 30/30
- Computer Security (Gabriele D'Angelo), 30L/30
- Data Base Systems (Matteo Golfarelli), 30/30
- Distributed Systems (Andrea Omicini), 30/30
- Engineering Complex Adaptive Software Systems (Mirko Viroli), 30/30
- Programming Languages and Models of Computation (Gianluigi Zavattaro), 30/30
- Project Management (Marco Antonio Boschetti), 30L/30
- Semantic Web (Antonella Carbonaro), 30L/30
- Software Systems Engineering (Antonio Natali), 30/30
- Web Services and Applications (Mario Bravetti), 30L/30

Bachelor's Degree in Electronics, Informatics, and Telecommunications Engineering

2009/09 → 2013/03

Università di Bologna (IT)

Grade Summa Cum Laude (Grade Average: 29.29/30, 5 laudes)

Thesis [Reuse Mechanisms and Concurrency: from Actors to Agent-oriented Programming](#) (supervisor: Alessandro Ricci)

Exams

- Automatic Controls (Paolo Castaldi), 30L/30
- Computer Networks (Claudio Salati), 30L/30
- Data Base Systems (Alessandra Lumini), 30L/30
- Digital Design Principles and Computer Architecture (Luca Roffia), 30/30
- Economics and Business Organisation (Cinzia Daraio), 28/30
- Electrotechnics (Franco Mastri), 28/30
- Foundations of Informatics A (Mirko Viroli), 30/30
- Foundations of Informatics B (Andrea Roli), 30/30
- General Physics A (Maurizio Piccinini), 28/30
- General Physics B (Maurizio Piccinini), 26/30
- Geometry and Algebra (Michele Mulazzani), 30/30
- Mathematical Analysis A (Massimo Cicognani), 28/30
- Mathematical Analysis for the Engineering Information Technology (Massimo Cicognani), 28/30
- Operating Systems (Alessandro Ricci), 29/30
- Operations Research (Daniele Vigo), 30L/30
- Signal Processing (Davide Dardari), 28/30
- Software Engineering (Antonio Natali), 30L/30
- Telecommunications Networks (Franco Callegati), 30L/30
- Web-related Technologies (Mario Bravetti), 30/30
- English Proficiency B1

Erasmus Programme

2012/01 → 2012/05

Courses

- Distributed Systems (Reiner Dojen), A/A
- Human-Computer Interaction (Luigina Cioffi), A/A
- Real-time Systems (Brian Adley), A/A
- Software Testing and Inspection (Norah Power), A/A

University of Limerick (IRL)

Skills and Technical Expertise

Note: this section is only indicative; current level of mastery can vary; by no means exhaustive.

Paradigms Imperative; OOP; Functional; Reactive; Async; Logic; Agent-Oriented

Languages Scala, Java/Kotlin, C#, C++, Ruby, Haskell, Python, JavaScript

Data E/R modeling; relational modelling; semantic web

Design/Arch. Design patterns; SOA/Microservices; cloud-native applications

Devops Docker; Kubernetes; CI/CD (Gradle, Travis, GitHub Actions)

Technologies **Cloud** – Google Cloud Platform, Amazon Web Services, Heroku
Web dev. – HTML5; CSS3; jQuery; PHP
Frameworks – Akka; Spring; NodeJs; ASP.NET MVC; Rails; Vert.x; RabbitMQ
Data – MySQL; NoSQL (e.g. MongoDB)

Process **Agile sw dev. and practices:** Scrum; (A)TDD/BDD.
Model-driven sw dev.: UML; DSL; code-generation (XText).
Collaborative sw dev.: version control (git); build automation (Gradle, sbt).

Languages

Italian Mother tongue.

English Proficient in both spoken and written English.

Referees

Prof. Mirko Viroli, Full Professor, Department of Computer Science and Engineering (DISI), Alma Mater Studiorum–Università di Bologna

Prof. Viroli was my early research and PhD advisor.

Referees from academia

Other referees for my academic profile include: Prof. Simon Dobson, Prof. Ferruccio Damiani, Prof. Andrea Omicini, Prof. Lukas Esterle, Prof. Jacob Beal, Prof. Giancarlo Fortino, Prof. Danny Weyns, Prof. Schahram Dustdar, Prof. Franco Zambonelli.

I hereby authorize the use of my personal data in accordance to the GDPR (General Data Protection Regulation) 679/16 – “European regulation on the protection of personal data”.